

Achieve quota, population monotonicity, and ex-ante proportionality without varying house size

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Website #33 | Solver verdict: DISPROVED | Verifier verdict: ISSUES | Shown: solver output

Problem

For an apportionment instance (p, H) with $p = (p_1, \dots, p_n) \in \mathbb{N}^n$, $H \in \mathbb{N}$, total population $P = \sum_i p_i$, and quotas

$$q_i = \frac{p_i}{P} H,$$

we seek a randomized method that outputs an integer vector $X = (X_1, \dots, X_n) \in \mathbb{N}_0^n$ such that $\sum_i X_i = H$ almost surely, and simultaneously satisfies:

1. *Quota compliance*: $X_i \in \{\lfloor q_i \rfloor, \lceil q_i \rceil\}$ almost surely for every i .
2. *Strong population monotonicity*: whenever two instances (p, H) and (p', H') satisfy

$$\frac{p'_i}{p'_j} \geq \frac{p_i}{p_j},$$

it must not happen that state i loses a seat while state j gains one.

3. *Ex-ante proportionality*: $\mathbf{E}[X_i] = q_i$ for every i .

I will prove that, under the strongest natural ex-post reading of randomized monotonicity, *no such method exists*. In fact, quota compliance and strong population monotonicity are already incompatible; ex-ante proportionality is irrelevant to the impossibility.

Alternative readings considered

The key ambiguity is what “population monotonicity” means for a randomized method across *different* instances.

1. **Chosen reading (hardest/natural)**. A randomized method is a probability distribution over deterministic solution maps f ; equivalently, one random seed ω fixes a deterministic apportionment rule f_ω on *all* instances. Quota and population monotonicity are then required to hold for almost every realized deterministic rule f_ω . This is the ex-post/support-wise reading used in recent work formalizing randomized apportionment. (csd.cmu.edu)
2. **Weaker marginal-only readings**. If one asks only for monotonicity of expectations, then ex-ante proportionality already makes first-order monotonicity essentially trivial, because the expectation equals quota. That is too weak to be the intended challenge.
3. **Intermediate pairwise-coupling readings**. One could ask only for special couplings between the two marginal laws associated with two instances. I do not adopt this weaker reading, because the prompt asks for the strongest standard monotonicity requirement, and the ex-post/global-seed reading is the mathematically natural one for comparing realized outcomes across instances.

Related literature found

- **Grimmett (2004) — BACKGROUND.** Introduced stochastic apportionment with fixed house size, quota ex post, and exact proportionality in expectation. This shows that quota + ex-ante proportionality is easy at a *single* instance; the difficulty is cross-instance monotonicity. (tandfonline.com)
- **Gölz, Peters, Procaccia (2025) — PARTIAL.** Gives a fixed-house randomized method satisfying quota, ex-ante proportionality, and ex-post *house* monotonicity, but not population monotonicity. So fixed house size is compatible with quota for a weaker monotonicity axiom. (pubsonline.informs.org)
- **Correa, Gölz, Schmidt-Kraepelin, Tucker-Foltz, Verdugo (2024) — PARTIAL.** Studies higher-order stochastic monotonicity notions (selection monotonicity, threshold monotonicity) for randomized rounding rules; this is a different notion from ex-post pairwise population monotonicity. (ar5iv.org)
- **Cembrano, Correa, Schmidt-Kraepelin, Tsigonias-Dimitriadis, Verdugo (2024, arXiv) — PARTIAL.** Explicitly relaxes fixed ex-post house size and, in that variable-house model, obtains population monotonicity, quota, and ex-ante proportionality together. This is the closest positive near-neighbor to the present problem. (arxiv.org)
- **Gölz (2022, PhD thesis) — SOLVES.** Chapter 8 states essentially the same negative result under the ex-post/support-wise reading: quota and population monotonicity are incompatible even before imposing ex-ante proportionality. I use this source only as orientation; the proof below is independent and self-contained. (csd.cmu.edu)

Proof sketch

Fix a base five-state profile

$$A = (824, 44, 44, 44, 44), \quad H = 10.$$

Its quotas are

$$(8.24, 0.44, 0.44, 0.44, 0.44),$$

so quota forces state 1 to receive either 8 or 9 seats.

Now split into the two possible quota-compliant cases.

If state 1 gets 9 seats on A , then exactly one small state gets the remaining seat. Inflate *that very state* to population 222, leaving the other three small states unchanged. Quota on the new profile forces state 1 down to at most 7 seats and the inflated state to at most 2, so at least one of the three unchanged small states must gain a seat. Thus an unchanged state gains while state 1, whose population is also unchanged relative to that state, loses. This violates strong population monotonicity because the relevant population ratio stayed constant.

If state 1 gets 8 seats on A , then exactly two small states get one seat each. Keep exactly those two winners unchanged and shrink the other two small states from 44 to 1. Quota on the new profile forces state 1 up to at least 9 seats, leaving room for at most one small-state seat in total. Hence at least one of the two unchanged previous winners must lose its seat, while state 1 gains. Again the relevant population ratio is unchanged, so strong population monotonicity is violated.

This yields a finite deterministic obstruction. The randomized impossibility then follows immediately: under the ex-post/global-seed reading, an almost-surely quota-compliant and population-monotone randomized method would, with probability 1, realize a deterministic rule satisfying these finite constraints, contradicting the obstruction.

Main result

Theorem 1. *Under the ex-post/global-seed interpretation of randomized apportionment, there is no randomized apportionment method with fixed house size H almost surely that satisfies quota compliance and strong pairwise-ratio population monotonicity for every instance. Consequently, there is no method satisfying conditions (1), (2), and (3) of the problem statement.*

In fact, the impossibility already holds without condition (3): quota compliance and strong population monotonicity alone are incompatible.

Proof

Lemma 1. *Let f be a deterministic apportionment rule satisfying strong pairwise-ratio population monotonicity. Suppose two instances (p, H) and (p', H') and two states $i \neq j$ satisfy*

$$p'_i = p_i, \quad p'_j = p_j.$$

Then it is impossible that

$$f_i(p', H') < f_i(p, H) \quad \text{and} \quad f_j(p', H') > f_j(p, H).$$

Proof. Because $p'_i = p_i$ and $p'_j = p_j$, we have

$$\frac{p'_i}{p'_j} = \frac{p_i}{p_j}.$$

Hence the ratio weakly increases in favor of state i (indeed it is unchanged). Strong pairwise-ratio population monotonicity therefore forbids state i from losing a seat while state j gains one. $\square$

Lemma 2 (Finite obstruction). *There is no deterministic apportionment rule f on the following finite family of instances that is both quota-compliant and strong pairwise-ratio population-monotone.*

All instances have house size $H = 10$. Define:

$$A = (824, 44, 44, 44, 44).$$

For each $k \in \{2, 3, 4, 5\}$, let

$$B_1^k = 824, \quad B_k^k = 222, \quad B_r^k = 44 \quad (r \notin \{1, k\}).$$

For each two-element set $\{a, b\} \subseteq \{2, 3, 4, 5\}$, let C^{ab} be defined by

$$C_1^{ab} = 824, \quad C_a^{ab} = C_b^{ab} = 44,$$

and for the complementary two states $r, s \in \{2, 3, 4, 5\} \setminus \{a, b\}$,

$$C_r^{ab} = C_s^{ab} = 1.$$

Proof. Assume for contradiction that such an f exists.

For the base instance A , the total population is 1000, so the quotas are

$$q(A) = (8.24, 0.44, 0.44, 0.44, 0.44).$$

By quota compliance,

$$f_1(A) \in \{8, 9\}.$$

Case 1: $f_1(A) = 9$.

Then exactly one of the four small states receives the remaining seat. Let $k \in \{2, 3, 4, 5\}$ be that unique state, so

$$f_k(A) = 1, \quad f_r(A) = 0 \quad \text{for all } r \in \{2, 3, 4, 5\} \setminus \{k\}.$$

Now consider the instance B^k . Its total population is

$$824 + 222 + 44 + 44 + 44 = 1178.$$

Therefore

$$q_1(B^k) = \frac{8240}{1178} = \frac{4120}{589} < 7,$$

so quota compliance gives

$$f_1(B^k) \leq 7.$$

Also

$$q_k(B^k) = \frac{2220}{1178} = \frac{1110}{589} < 2,$$

so

$$f_k(B^k) \leq 2.$$

For each $r \in \{2, 3, 4, 5\} \setminus \{k\}$,

$$q_r(B^k) = \frac{440}{1178} = \frac{220}{589} < 1,$$

hence $f_r(B^k) \in \{0, 1\}$.

Since the total number of seats is 10, we have

$$\sum_{r \in \{2, 3, 4, 5\} \setminus \{k\}} f_r(B^k) = 10 - f_1(B^k) - f_k(B^k) \geq 10 - 7 - 2 = 1.$$

Thus there exists some $r \in \{2, 3, 4, 5\} \setminus \{k\}$ with

$$f_r(B^k) = 1.$$

But $f_r(A) = 0$, while $f_1(A) = 9$ and $f_1(B^k) \leq 7$. So between A and B^k ,

$$f_1(B^k) < f_1(A), \quad f_r(B^k) > f_r(A).$$

Moreover, the populations of states 1 and r are unchanged:

$$A_1 = B_1^k = 824, \quad A_r = B_r^k = 44.$$

By the previous lemma, this is impossible. Contradiction.

Case 2: $f_1(A) = 8$.

Then the remaining two seats go to exactly two small states. Let $\{a, b\} \subseteq \{2, 3, 4, 5\}$ be the two winners, so

$$f_a(A) = f_b(A) = 1,$$

and the other two small states get 0 seats.

Now consider C^{ab} . Its total population is

$$824 + 44 + 44 + 1 + 1 = 914.$$

Hence

$$q_1(C^{ab}) = \frac{8240}{914} = \frac{4120}{457} > 9,$$

so quota compliance gives

$$f_1(C^{ab}) \geq 9.$$

Also

$$q_a(C^{ab}) = q_b(C^{ab}) = \frac{440}{914} = \frac{220}{457} < 1,$$

thus

$$f_a(C^{ab}) \leq 1, \quad f_b(C^{ab}) \leq 1.$$

Because $f_1(C^{ab}) \geq 9$ and the house size is 10, at most one of the four small states can receive a seat in C^{ab} . Therefore at least one of a, b must lose its seat. Choose $t \in \{a, b\}$ such that

$$f_t(C^{ab}) = 0.$$

Then

$$f_t(C^{ab}) < f_t(A), \quad f_1(C^{ab}) > f_1(A),$$

since $f_t(A) = 1$, $f_1(A) = 8$, and $f_1(C^{ab}) \geq 9$.

Again, the populations of states t and 1 are unchanged:

$$C_t^{ab} = A_t = 44, \quad C_1^{ab} = A_1 = 824.$$

So by the previous lemma this is impossible. Contradiction.

Both cases lead to contradictions. Therefore no such deterministic f exists. $\square$

Proof of the theorem. Assume, for contradiction, that a randomized method M with the required properties exists under the chosen ex-post/global-seed reading.

Thus there is a probability space $(\Omega, \mathcal{F}, \mathbf{P})$ such that each $\omega \in \Omega$ determines a deterministic apportionment rule f_ω . By assumption, quota compliance and strong population monotonicity hold almost surely.

Consider only the finite set of instances

$$\mathcal{I} = \{A\} \cup \{B^k : k = 2, 3, 4, 5\} \cup \{C^{ab} : \{a, b\} \subseteq \{2, 3, 4, 5\}, |\{a, b\}| = 2\}.$$

For each profile in $\mathcal{I}$, quota compliance is an almost-sure event. For each comparison used in the finite obstruction lemma, the relevant strong population monotonicity requirement is also an almost-sure event. Since there are only finitely many such events, their intersection still has probability 1.

Choose ω in that intersection. Then the realized deterministic rule f_ω , restricted to $\mathcal{I}$, is quota-compliant and strong pairwise-ratio population-monotone. This contradicts the finite obstruction lemma.

Therefore no such randomized method exists.

Finally, note that ex-ante proportionality was never used in the contradiction. So the impossibility is stronger: quota compliance and strong population monotonicity alone already rule out existence. $\square$

Gap to the full statement

None for the chosen ex-post reading. The theorem outright disproves existence.

If one weakens the meaning of randomized monotonicity to a purely marginal or expectation-based condition, one obtains a different problem; I did not analyze that weaker variant because the prompt asked for the strongest natural reading.

Self red-team

1. Counterexample attempts against the proof.

- I checked the critical quota inequalities explicitly:

$$\frac{8240}{1178} < 7, \quad \frac{2220}{1178} < 2, \quad \frac{8240}{914} > 9.$$

These are the only nontrivial numerical facts the argument uses.

- I checked that each contradiction uses states whose populations are unchanged between the two compared instances, so the ratio condition is satisfied with equality, which is enough for the “weakly increases” hypothesis.
- I checked that no hidden symmetry or anonymity assumption is used. The proof never says “without loss of generality” about which small state got the seat(s): the comparison profiles B^k and C^{ab} are indexed by the *actual* winner(s) in the base instance.

2. Three likely referee objections.

- *Objection: Randomization might evade the deterministic obstruction.* Answer: under the chosen ex-post reading, randomization is a distribution over deterministic rules. Because the obstruction uses only finitely many instances and finitely many monotonicity constraints, an almost-sure method would realize a deterministic rule satisfying all of them on a probability-one event.
- *Objection: The prompt uses strong pairwise-ratio monotonicity, but the proof only compares unchanged states.* Answer: unchanged states are enough, because equality of ratios is a special case of “weakly increases.”
- *Objection: Perhaps the contradiction only rules out ex-ante proportionality.* Answer: no. Ex-ante proportionality is not used anywhere. The contradiction is already between quota and ex-post strong population monotonicity.

3. Boundary and degenerate checks.

- All populations are positive integers, so no zero-population edge case is involved.
- All compared instances use the same house size $H = 10$, so the contradiction is not smuggling in house-monotonicity issues.
- The proof does *not* accidentally prove anything false about house monotonicity: it never compares different house sizes.

4. Compression/minimality search.

I tried to compress the obstruction to four states by keeping one large state and only three small states. Variants of the above two-branch argument could make one branch work, but I could not make both branches close simultaneously while maintaining the needed quota inequalities. So the present proof shows that $n = 5$ suffices; I do not claim minimality.

Invoked results

No external theorem is invoked as a proof step. The proof above is self-contained. The cited literature is contextual only, so there is no citation-dependent hypothesis audit to perform inside the proof.

Self-confidence

3 — The finite obstruction and the reduction from randomized to deterministic realizations are fully checked; the only caveat is that the prompt admits weaker stochastic interpretations, which I explicitly rejected in favor of the strongest natural one.

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